

Exception Handling Specification

Smart Lab Equipment & Slot Reservation Portal

Exceptions below are derived from the project's stated reservation, calibration, double-booking, authentication, concurrent-processing, and late-return requirements.

ID	Exception / Condition	System Response	Result
E-01	Invalid or expired equipment calibration.	Verify calibration before confirmation. Do not confirm the reservation; require the student to select calibrated equipment.	No reservation for uncalibrated equipment.
E-02	Selected equipment/time slot is already reserved.	Reject the conflicting request and prevent overlapping reservations for the same equipment.	No double-booking is created.
E-03	Requested slot exceeds the maximum 2-hour duration.	Reject the request and require a valid slot of no more than 2 hours.	Only a valid-duration reservation can proceed.
E-04	Requested slot is more than 7 days in advance.	Reject the request because it falls outside the allowed advance-reservation window.	Only reservations within 7 days can proceed.
E-05	User is unauthenticated or unauthorized to reserve equipment.	Deny the reservation request and do not create a reservation.	Unauthorized reservation cannot be created.
E-06	Concurrent requests target the same equipment and overlapping slot.	Use strict database-level lock isolation so concurrent requests cannot create a race-condition conflict.	Conflicting reservations are prevented.
E-07	Equipment is returned after the reserved end time.	Record the return information and identify the return as late for the applicable disciplinary process.	Late return is correctly identified.

Connection to UC-02 — Reserve Equipment Slot

The selected alternate flow is **A1 — Equipment/Slot Unavailable**. It corresponds to **E-02** above. The UML may represent this conditional behavior with an «extend» relationship from **Handle Unavailable Slot** toward **Reserve Equipment Slot**.

Traceability

Requirement	Exception
FR-001 — availability and reservation limits	E-02, E-03, E-04
FR-002 — calibration verification	E-01
FR-003 — prevent overlapping reservations	E-02, E-06
FR-004 — identify late returns	E-07
NFR-001 — concurrent reservation reliability	E-06
NFR-002 — authenticated/authorized reservation	E-05